

Differentiable Inverse Reachability for RM75: Complete Mathematical Design of the IRDPLAYGROUND Operator

IRDPLAYGROUND technical note

Abstract

This note gives a self-contained mathematical account of the inverse reachability (IRD) operator implemented in `ird_playground`. The operator answers a narrower question than classical discrete IRD tables: given a medically valid tool-centre-point (TCP) pose and a rail-induced robot base (J1-axis) frame, what smooth direction increases collision-checked arm reachability, optionally under a local position/orientation region? We relate the design to four foundational papers (Vahrenkamp et al. 2013; Rudorfer 2024/RM4D; Murooka et al. 2025; Chiu 1988), derive every aggregation formula used in code, and explain why automatic differentiation yields usable IRD gradients, why conditional queries over local pose cones are well-defined, and why the same field supports trajectory-parameter updates without retraining.

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1 Problem statement and classical IRD

1.1 Reachability and inverse reachability

Let the robot configuration be $q \in \mathcal{Q} \subset \mathbb{R}^n$ and let forward kinematics map configurations to TCP poses

$$\text{FK} : \mathcal{Q} \rightarrow \text{SE}(3), \quad q \mapsto T_{\text{tcp}}(q). \quad (1)$$

Here $\text{SE}(3)$ is the special Euclidean group of rigid poses (rotation $R \in \text{SO}(3)$ and translation $p \in \mathbb{R}^3$).

Definition 1 (Binary reachability). *A pose $r \in \text{SE}(3)$ is reachable if there exists a collision-free configuration q with $\text{FK}(q) = r$ (within IK tolerance) and joint limits satisfied. Write*

$$\mathcal{R} = \{ r \in \text{SE}(3) \mid \exists q \in \mathcal{Q}_{\text{free}} : \text{FK}(q) = r \}. \quad (2)$$

A classical *reachability distribution* (RD) of Vahrenkamp et al. (2013) stores, in a voxelisation of workspace relative to a fixed robot base, a quality score such as the measure of joint configurations that map into that voxel. An *inverse reachability distribution* (IRD) inverts the same data: given a fixed TCP (e.g. a grasp), it yields a distribution over candidate base poses for which that TCP would be reachable.

Formally, if $t_i = T_{\text{tcp}_i}^{-1}$ denotes a base-to-TCP sample stored in the RD, then for a world TCP $p \in \text{SE}(3)$ the corresponding candidate base pose is

$$b_i(p) = p t_i^{-1} \in \text{SE}(3) \quad (3)$$

(with optional projection onto $\text{SE}(2)$ for planar base placement). The IRD score of $b_i(p)$ is inherited from the RD quality of t_i . This is an offline discrete table lookup.

1.2 What IRDPLAYGROUND computes instead

Our first-stage operator does *not* replace multi-seed IK or final collision validation. It answers:

For a medically valid TCP pose T_{tcp} and a rail-induced J1-axis frame T_{axis} , what smooth direction increases collision-checked arm reachability clearance, including under a local position/orientation uncertainty or task region?

Denote the decision variables of interest by x (rail coordinate, TCP yaw on a vessel surface, trajectory control points, ...). We construct a scalar

$$C(x) = \mathcal{A}[f(c(T_{\text{tcp}}(x), T_{\text{axis}}(x)))] \quad (4)$$

where:

- $c(\cdot)$ is a smooth RM4D-style chart (Section 3);
- f is a learned signed clearance field (Section 4);
- $\mathcal{A}$ is a differentiable set aggregator (Region A, TaskCone, SetQuery, or Trajectory; Sections 6–8).

Gradient-based updates use $\nabla_x C$ from automatic differentiation through the entire Torch graph.

Paper	Core idea absorbed
Vahrenkamp et al. 2013	IRD = fix TCP, score base poses
Rudorfer RM4D 2024	Quotient common yaw/roll symmetries $\rightarrow \sim 4D$ map shared by RM & IRD
Murooka et al. 2025	Differentiable reachability map $f_R(r)$ for optimisation; classifiers can have flat exteriors
Chiu 1988	Postures should match task transmission requirements

2 Paper map: what each reference contributes

Murooka’s differentiable reachability map. Murooka et al. define a scalar $f_R(r)$ with

$$f_R(r) \geq 0 \text{ if reachable,} \quad f_R(r) < 0 \text{ otherwise,} \quad (5)$$

trained as a smooth binary classifier (e.g. softplus logistic loss). When f_R is continuous and differentiable w.r.t. task coordinates, the gradient $\partial f_R / \partial r$ can enter trajectory optimisation without repeated IK. They also note that a pure decision function may become nearly constant far outside the reachable set, yielding uninformative gradients. Our design follows the same sign convention but supervises an explicit signed clearance with local collision-checked stencils and boundary normals, so that ∇f remains informative near the frontier (Section 5).

RM4D symmetry. Rudorfer observes that for many 6-/7-axis arms, reachability is approximately invariant to base yaw about gravity and to TCP roll about the last joint, reducing an SE(3) query to an intrinsic 4-D quotient. We adopt the dimensionality-reduction spirit but enforce only the symmetries that hold for RM75 with the probe45 tool: base (J1) yaw is a true free symmetry; TCP/flange roll is *not*, because the TCP z -axis is offset by $\sim 50^\circ$ from the J7 axis.

Chiu’s task compatibility. Chiu measures how well a posture’s velocity/force ellipsoids align with a task. We do not compute Yoshikawa indices online; instead we encode task freedom (large tip $\times$ roll cone) and medical uncertainty (tiny box $\times$ cone) as set queries over f , so that C is a task-conditioned reachability margin.

3 Frames, rail base, and the flange chart

3.1 World poses and relative pose

A homogeneous pose is written

$$T = \begin{bmatrix} R & p \\ 0^\top & 1 \end{bmatrix} \in \text{SE}(3), \quad R \in \text{SO}(3), \quad p \in \mathbb{R}^3. \quad (6)$$

Given world TCP T_{tcp}^W and world J1-axis frame T_{axis}^W , the TCP expressed in the axis frame is

$$\begin{aligned} R_{\text{axis} \leftarrow \text{tcp}} &= (R_{\text{axis}}^W)^\top R_{\text{tcp}}^W, \\ p_{\text{axis} \leftarrow \text{tcp}} &= (R_{\text{axis}}^W)^\top (p_{\text{tcp}}^W - p_{\text{axis}}^W). \end{aligned} \quad (7)$$

Interpretation. Equation (7) is the group product $(T_{\text{axis}}^W)^{-1} T_{\text{tcp}}^W$. All reachability queries are posed in this relative frame, which is exactly the IRD viewpoint: move the base/axis and the relative TCP changes even if the world TCP is fixed.

3.2 Rail-induced axis frame

For a linear rail along local axis e_a (code default $a = 1$, i.e. y),

$$T_{\text{axis}}^W(y) = T_{\text{world} \leftarrow \text{rail}} \underbrace{\begin{bmatrix} I_3 & y e_a \\ 0^\top & 1 \end{bmatrix}}_{\text{pure translation by rail coordinate } y} T_{\text{rail} \leftarrow \text{axis}}(y=0). \quad (8)$$

Interpretation. y is the raw URDF rail coordinate. Differentiating (8) w.r.t. y moves only the axis origin along the rail; rotation is constant. This is the primitive that turns “IRD w.r.t. base placement” into “IRD w.r.t. one scalar rail DoF”.

3.3 TCP to flange

Let $T_{\text{flange} \leftarrow \text{tcp}}$ be the fixed tool transform from URDF. Flange (link_7) pose in the axis frame is

$$(p_{\text{fl}}, R_{\text{fl}}) = \text{compose}((p_{\text{tcp}}, R_{\text{tcp}}), T_{\text{flange} \leftarrow \text{tcp}}). \quad (9)$$

We score flange geometry rather than TCP geometry so that the chart axes align with the J7 rotation axis (needed for the invariant / roll split below).

3.4 Nine-dimensional yaw-invariant flange embedding

Write flange position $p = (p_x, p_y, p_z)$ and flange axes u_x, u_y, u_z (columns of R_{fl}). Define the radial smoothed distance

$$r = \sqrt{p_x^2 + p_y^2} + \varepsilon, \quad \varepsilon = (10^{-3} \text{ m})^2. \quad (10)$$

The chart is the map

$$c(p, R) = \begin{pmatrix} p_z \\ r \\ u_x \cdot \hat{z} \\ p_{xy} \cdot u_{x,xy} \\ p_{xy} \times u_{x,xy} \\ u_z \cdot \hat{z} \\ p_{xy} \cdot u_{z,xy} \\ p_{xy} \times u_{z,xy} \\ u_y \cdot \hat{z} \end{pmatrix} \in \mathbb{R}^9, \quad (11)$$

where $p_{xy} = (p_x, p_y)$, $u_{\cdot,xy}$ is the xy -projection of an axis, $\hat{z} = (0, 0, 1)$, the 2-D “cross” is the scalar $p_x u_y - p_y u_x$, and

$$u_y \cdot \hat{z} = (u_z \times u_x) \cdot \hat{z} = u_{z,x} u_{x,y} - u_{z,y} u_{x,x}. \quad (12)$$

Why each component.

- p_z : height along the J1 axis (invariant to yaw about $\hat{z}$).
- r : cylindrical radius; ε removes the kink of $\sqrt{p_x^2 + p_y^2}$ at the axis (curvature $\sim 10^3/\text{m}$ instead of $\sim 10^6/\text{m}$).
- Components involving u_z : J7-invariant block (indices $\{0, 1, 5, 6, 7\}$ in code), because spinning about J7 rotates u_x, u_y but not u_z .
- Components involving u_x and $u_y \cdot \hat{z}$: *roll-carrying* block. They retain flange roll γ , which is a genuine DoF for probe45 and must not be quotiented.
- The ninth entry $u_y \cdot \hat{z}$ completes $R^\top \hat{z}$. Without it, as $r \rightarrow 0$ the mixed products vanish and a twofold ambiguity remains in reconstructing orientation about the axis.

Symmetry statement. If $R_z(\alpha)$ is a rotation about the J1 axis applied simultaneously to base and (equivalently) to the relative flange pose, then

$$c(R_z(\alpha)p, R_z(\alpha)R) = c(p, R). \quad (13)$$

Thus c embeds the intrinsic quotient $\text{SE}(3)/\text{Yaw}_{J1}$ smoothly into $\mathbb{R}^9$ (redundant coordinates of an intrinsic 5-D flange chart after quotienting only yaw).

Remark 1 (Legacy 5-D TCP chart). *An older embedding $c_{\text{TCP}} = [p_z, u_z \hat{z}, \|p_{xy}\|, p_{xy} u_{xy}, p_{xy} \times u_{xy}]$ quotiented both yaw and TCP roll. For probe45 this is incorrect and is kept only for audits.*

4 Signed reachability field

4.1 Definition

Definition 2 (Signed clearance field). *The network $f_\theta : \mathbb{R}^9 \rightarrow \mathbb{R}$ approximates a signed reachability clearance on the flange chart:*

$$f_\theta(c) \begin{cases} > 0 & \text{reachable (positive margin),} \\ = 0 & \text{approximate boundary,} \\ < 0 & \text{unreachable.} \end{cases} \quad (14)$$

This matches Murooka’s sign pattern (5) but the *magnitude* is trained to track a collision-checked local distance in a declared metric (Section 5.1), not merely a classification margin.

4.2 Input normalisation and Fourier features

Training chart rows determine centre $\mu \in \mathbb{R}^9$ and scale $\sigma \in \mathbb{R}^9$ (quantile mid-range). Normalised inputs are

$$\tilde{c} = \frac{c - \mu}{\sigma} \quad (\text{elementwise}). \quad (15)$$

Fourier features with B bands ($B=3$ by default) are

$$\Phi(\tilde{c}) = [\tilde{c}, \sin(\pi 2^0 \tilde{c}), \cos(\pi 2^0 \tilde{c}), \dots, \sin(\pi 2^{B-1} \tilde{c}), \cos(\pi 2^{B-1} \tilde{c})] \in \mathbb{R}^{9(1+2B)}. \quad (16)$$

Interpretation. Metres and dimensionless orientation channels have incompatible dynamic ranges; without (15), $\sin(\pi 2^k x)$ would scramble them. Fourier lifting helps represent sharp reachable/unreachable transitions without a spatial hash grid (the previous model memorised boundary samples).

4.3 Network architecture

Let $\text{softplus}_\beta(z) = \beta^{-1} \log(1 + e^{\beta z})$. The field is a residual MLP with Softplus activations and a linear scalar head:

$$\begin{aligned} h_0 &= \text{softplus}_\beta(W_0 \Phi(\tilde{c}) + b_0), \\ h_{\ell+1} &= \text{softplus}_\beta(h_\ell + W_\ell^{(2)} \text{softplus}_\beta(W_\ell^{(1)} h_\ell + b_\ell^{(1)}) + b_\ell^{(2)}), \\ f_\theta(c) &= w^\top h_L + b. \end{aligned} \quad (17)$$

Softplus (unlike ReLU) is C^∞ with nowhere-vanishing derivative for finite inputs, which is desirable when $\nabla_c f$ is used as a guidance direction.

4.4 Support guard

Let $[\underline{c}, \bar{c}]$ be the axis-aligned box of training chart rows. After normalisation, a smooth exterior distance is

$$d_{\text{supp}}(c) = \sqrt{\|\text{ReLU}(\underline{c} - \tilde{c}) + \text{ReLU}(\tilde{c} - \bar{c})\|_2^2 + \epsilon^2} - \epsilon, \quad (18)$$

and the guarded score is

$$\tilde{f}(c) = f_\theta(c) - \lambda_g d_{\text{supp}}(c), \quad \lambda_g = 8 \text{ (default)}. \quad (19)$$

Interpretation. Outside the training support, $\tilde{f}$ is pulled negative so optimisation is discouraged from extrapolating into unlabelled chart regions.

4.5 World scoring (the IRD point query)

$$F(T_{\text{tcp}}^W, T_{\text{axis}}^W) = \tilde{f}\left(c(\text{flange}((T_{\text{axis}}^W)^{-1}T_{\text{tcp}}^W))\right). \quad (20)$$

Interpretation. Fix T_{tcp}^W and vary $T_{\text{axis}}^W(y)$ via (8): then F is precisely a smooth IRD score of the rail/base placement. Fix the axis and vary T_{tcp}^W : then F is a smooth forward reachability score. One field serves both views because of the relative chart (11).

5 Declared SE(3) metric and training targets

5.1 Metric

Boundary labels use the weighted distance that equates 1 cm of translation with 1° of rotation:

$$\lambda = \frac{0.01 \text{ m}}{1^\circ \text{ in radians}} = \frac{0.01}{\pi/180} \approx 0.5730 \text{ m/rad}. \quad (21)$$

For a translational increment $\Delta p \in \mathbb{R}^3$ (metres) and rotational increment $\Delta \omega \in \mathbb{R}^3$ (radians, rotation-vector / so(3) coordinates),

$$d(\Delta p, \Delta \omega) = \sqrt{\|\Delta p\|_2^2 + \lambda^2 \|\Delta \omega\|_2^2}. \quad (22)$$

Interpretation. This is *not* the bi-invariant SE(3) geodesic; it is a declared chart-friendly metric used to place stencil samples and to interpret clearance in “metres”.

5.2 Stencil supervision

Around collision-checked boundary centres one evaluates exact reachable / unreachable labels at offsets

$$\pm 1, 3, 6, 10 \text{ mm} \quad \text{and} \quad \pm 1, 3, 5^\circ, \quad (23)$$

plus explicit zero-valued bisected boundary centres. These provide targets $s^*(c)$ for the signed value of $f_\theta(c)$.

5.3 Losses

Let $\{(c_i, y_i, s_i^*, n_i)\}_i$ be a minibatch with binary label $y_i \in \{0, 1\}$, SDF target s_i^* , and (when available) unit boundary normal n_i in normalised chart coordinates.

Classification (sign).

$$\mathcal{L}_{\text{cls}} = \frac{\sum_i w_i^{\text{cls}} \ell_{\text{BCE}}(\alpha f_\theta(c_i), y_i)}{\sum_i w_i^{\text{cls}}}, \quad (24)$$

where α is a logit scale (default 3) and ℓ_{BCE} is binary cross-entropy with logits. *Interpretation.* Enforces Murooka-style sign correctness (14).

Signed value (magnitude). On rows with SDF weight $w_i^{\text{sdf}} > 0$,

$$\mathcal{L}_{\text{sdf}} = \text{SmoothL1}_{\beta=0.1}(f_\theta(c_i), s_i^*). \quad (25)$$

Interpretation. Makes $|f|$ meaningful as clearance, not an arbitrary classifier margin.

Boundary normal alignment. With $\tilde{c}$ requiring grad,

$$\mathcal{L}_{\text{n}} = \mathbb{E}_i \left[1 - \frac{\nabla_{\tilde{c}} f_\theta(c_i) \cdot n_i}{\|\nabla_{\tilde{c}} f_\theta(c_i)\| \|n_i\|} \right]. \quad (26)$$

Interpretation. Cosine loss aligns the field gradient with the collision-checked crossing direction. This is why ∇F points toward increasing reachability near the frontier rather than along arbitrary level sets.

Optional Eikonal (disabled in production).

$$\mathcal{L}_{\text{Eik}} = \text{SmoothL1}(\|\nabla_{\tilde{c}} f\|, v^*), \quad (27)$$

with $v^* = 1$ or an empirical stencil slope. Workspace-normalised mixed units make a universal unit-speed prior poorly conditioned; production relies on (25)–(26) instead.

Total.

$$\mathcal{L} = \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{sdf}} \mathcal{L}_{\text{sdf}} + \lambda_{\text{n}} \mathcal{L}_{\text{n}} + \lambda_{\text{Eik}} \mathcal{L}_{\text{Eik}} + \dots \quad (28)$$

Remark 2 (Why not a classifier alone?). *A classifier can achieve high accuracy while being nearly constant for c deep in the unreachable region, so $\nabla_x F \approx 0$ exactly when an optimiser most needs a recovery direction (Murooka’s caveat). Signed stencil values plus normal alignment keep a usable slope across the boundary band.*

6 Differentiable set queries

All set queries share the pattern: sample poses about a nominal TCP, score each with (20), reduce with a smooth aggregator.

6.1 Local pose perturbation

In the medical TCP frame with columns $[b, t, n]$ (binormal, tangent, normal), a position offset $\delta p_{\text{loc}} \in \mathbb{R}^3$ and rotation vector $\delta \omega_{\text{loc}} \in \mathbb{R}^3$ produce

$$\begin{aligned} p' &= p + R \delta p_{\text{loc}}, \\ R' &= R \text{Exp}(\delta \omega_{\text{loc}}), \end{aligned} \quad (29)$$

where $\text{Exp} : \mathbb{R}^3 \rightarrow \text{SO}(3)$ is the Rodrigues map

$$\text{Exp}(\omega) = I + \frac{\sin \theta}{\theta} [\hat{\omega}] + \frac{1 - \cos \theta}{\theta^2} [\hat{\omega}]^2, \quad \theta = \|\omega\|, \quad (30)$$

and $[\hat{\omega}]$ is the skew-symmetric matrix of the unit axis (with the usual $\theta \rightarrow 0$ limit $I + [\omega]$).

6.2 Normalised softmin and softmax

For a vector $v \in \mathbb{R}^K$ and temperature $\tau > 0$,

$$\text{LogSumExp}(v) = \log \sum_{k=1}^K e^{v_k}. \quad (31)$$

The *normalised softmin* used for robust aggregation is

$$\text{softmin}_\tau(v) = -\tau \left(\text{LogSumExp}(-v/\tau) - \log K \right). \quad (32)$$

Why the $\log K$ term? If $v_k \equiv a$ for all k , then $\text{LogSumExp}(-v/\tau) = \log K - a/\tau$, hence $\text{softmin}_\tau(v) = a$. A constant field is unchanged; without $\log K$, softmin would shift by $-\tau \log K$.

The *normalised softmax-like maximum* is

$$\text{softmax}_\tau(v) = \tau \left(\text{LogSumExp}(v/\tau) - \log K \right). \quad (33)$$

Same constant-preserving property for maxima.

Implicit weights (gradient structure). Differentiating (32) gives

$$\frac{\partial \text{softmin}_\tau(v)}{\partial v_k} = \frac{e^{-v_k/\tau}}{\sum_j e^{-v_j/\tau}} = \pi_k^{\min}(v), \quad (34)$$

i.e. a Gibbs distribution concentrated on the *smallest* entries. For (33),

$$\frac{\partial \text{softmax}_\tau(v)}{\partial v_k} = \frac{e^{v_k/\tau}}{\sum_j e^{v_j/\tau}} = \pi_k^{\max}(v), \quad (35)$$

concentrated on the *largest* entries. Thus robust queries backpropagate into worst-case samples; task-free queries backpropagate into best-case samples.

6.3 Region A: uncertainty box $\times$ small cone (softmin)

Definition 3 (Region A). *Default extents: binormal 4 mm, tangent 3 mm, normal 2 mm; orientation cone half-angle $\beta = 3^\circ$; $K = 64$ fixed scrambled Sobol samples including the nominal pose.*

Position half-samples use a joint 5-D Sobol sequence $u \in (0, 1)^5$:

$$\delta p = (2u_{1:3} - 1) \odot (b, t, n)_{\text{extent}}, \quad (36)$$

and orientation samples are uniform in solid angle inside the cone,

$$\begin{aligned} \cos \rho &= 1 - u_4(1 - \cos \beta), \\ \rho &= \arccos(\cos \rho), \\ \phi &= 2\pi u_5, \\ \delta \omega &= (\rho \cos \phi, \rho \sin \phi, 0), \end{aligned} \quad (37)$$

then antisymmetrised $\{0, \pm \text{half samples}\}$ to length K . The Region A clearance is

$$C_A(T_{\text{tcp}}, T_{\text{axis}}) = \text{softmin}_\tau \left\{ F(T_{\text{tcp}}(\delta_k), T_{\text{axis}}) \right\}_{k=1}^K, \quad \tau = 0.15. \quad (38)$$

Interpretation. C_A is a smooth lower envelope over a medically motivated uncertainty set U . Increasing C_A improves the worst likely pose under small registration / contact error. Coverage diagnostics use $\frac{1}{K} \sum_k \sigma(F_k/\tau)$.

6.4 Task cone: tip $\times$ roll free set (softmax)

Ultrasound contact often permits a large tip tilt and probe roll as *controller-chosen* free variables, not as uncertainty.

Definition 4 (TaskCone). *Tip half-angle β_{tip} (e.g. 20° – 45°), roll half-range γ_{max} (e.g. 20° – 30°), K Sobol samples in (ρ, ϕ, γ) .*

$$\begin{aligned}\cos \rho &= 1 - u_1(1 - \cos \beta_{\text{tip}}), \\ \phi &= 2\pi u_2, \\ \gamma &= (2u_3 - 1)\gamma_{\text{max}}, \\ \delta\omega &= (\rho \cos \phi, \rho \sin \phi, \gamma).\end{aligned}\tag{39}$$

Position is held fixed (contact point prescribed). The task clearance is

$$C_{\text{task}}(T_{\text{tcp}}, T_{\text{axis}}) = \text{softmax}_\tau \left\{ F(T_{\text{tcp}}(\eta_k), T_{\text{axis}}) \right\}_{k=1}^K, \quad \tau = 0.20.\tag{40}$$

Interpretation. Opposite of Region A: C_{task} asks whether *some* tip $\times$ roll choice makes the arm comfortable. Gradients push the base/rail toward placements that admit a high-clearance free pose inside the cone. This is the Chiu-style “match the posture set to the task” idea, expressed as a reachability margin rather than an ellipsoid index.

6.5 Nested free $\times$ uncertain query (SetQuery)

The general Phase-4 semantic is

$$C_{\text{set}}(T) = \max_{\eta \in D_{\text{free}}} \underbrace{\text{CVaR}_\alpha \text{ or } \text{softmin}}_{\delta \in U} F(c(T(\eta, \delta))).\tag{41}$$

Batch scores have shape $[N, K_{\text{free}}, K_{\text{unc}}]$. Reduce the uncertain axis first, then take a smooth max over free variables.

Rockafellar–Uryasev lower-tail CVaR. For clearance values (higher better), lower-tail CVaR at level α is

$$\text{CVaR}_\alpha(X) = \min_{t \in \mathbb{R}} \left\{ t + \frac{1}{1 - \alpha} \mathbb{E}[(t - X)_+] \right\}.\tag{42}$$

Interpretation. t plays the role of a VaR threshold; the expectation penalises shortfalls below t . Empirically the optimum lies in the sample set, so code evaluates the objective at every sample value and takes a hard or soft min over those candidate t ’s. This is more statistically stable than a pure softmin when K_{unc} is moderate.

Implemented reduction.

$$\begin{aligned}g(\eta) &= \text{CVaR}_\alpha \{ F(T(\eta, \delta_j)) \}_j \quad \text{or} \quad \text{softmin}_\tau \{ \dots \}, \\ C_{\text{set}} &= \text{softmax}_\tau \{ g(\eta_i) \}_i.\end{aligned}\tag{43}$$

7 Why IRD gradients exist and are useful

7.1 Chain rule through the IRD point query

Consider rail coordinate y with fixed world TCP. Then

$$\frac{dF}{dy} = \frac{\partial \tilde{f}}{\partial c} \frac{\partial c}{\partial(p, R)} \frac{\partial(p, R)}{\partial T_{\text{axis}}} \frac{dT_{\text{axis}}}{dy}.\tag{44}$$

Every factor is implemented with Torch ops (Exp, matrix products, $\sqrt{\cdot + \varepsilon}$, Softplus MLP), so dF/dy is exact up to floating-point error. Empirical AD-vs-FD median relative error on rail and TCP- x is on the order of 10^{-4} .

Geometric reading. $\partial c/\partial(p, R)$ converts a spatial twist of the relative flange pose into chart coordinates. $\partial \hat{f}/\partial c$ is trained to align with the outward reachable normal near the boundary (Eq. (26)). Hence $dF/dy > 0$ means “moving the carriage in the $+y$ direction increases predicted clearance”.

7.2 Gradients of conditional cone queries

For Region A,

$$\nabla_x C_A = \sum_{k=1}^K \pi_k^{\min} \nabla_x F(T_{\text{tcp}}(\delta_k; x), T_{\text{axis}}(x)). \quad (45)$$

For TaskCone,

$$\nabla_x C_{\text{task}} = \sum_{k=1}^K \pi_k^{\max} \nabla_x F(T_{\text{tcp}}(\eta_k; x), T_{\text{axis}}(x)). \quad (46)$$

Interpretation of “conditional IRD gradient”. The condition is the set U or D_{free} . The scalar C is the set-aggregated clearance. Its gradient w.r.t. base/rail/path parameters is a *weighted average of point-IRD gradients* at the sampled local poses, with weights (34) or (35). No discrete voxel argmax is required.

7.3 Vessel-surface example (θ, y)

On an elliptical vessel, TCP position is tied to surface angle θ and path abscissa s , while rail y sets the base. Demo code forms

$$C(s; \theta, y) = C_{\text{task}}(T_{\text{tcp}}(\theta, s), T_{\text{axis}}(y)) \quad (47)$$

and extracts

$$\frac{\partial C}{\partial \theta}, \quad \frac{\partial C}{\partial y} \quad (48)$$

by `torch.autograd.grad`. Quiver plots of these components are literal visualisations of the conditional IRD vector field on the (θ, y) planning plane.

8 Trajectory operator and online updates

8.1 Aggregation order

Given waypoints $\{T_{\text{tcp}}^{(\ell)}\}_{\ell=1}^W$ and a (shared or per-waypoint) axis frame, the trajectory operator applies, in order:

1. **Uncertainty / Region A scenarios** for each approximate-angle candidate;
2. **Angle aggregation** (softmax select, softmax robust, mean, or CVaR);
3. **Path residuals**: interior SE(3) samples on each segment;
4. **Trajectory reduction**: softmax / CVaR / hard min over waypoint and path margins.

8.2 SE(3) path interpolation

On a segment from T_0 to T_1 , for $\alpha \in (0, 1)$,

$$\begin{aligned} p(\alpha) &= (1 - \alpha)p_0 + \alpha p_1, \\ R(\alpha) &= R_0 \text{Exp}(\alpha \text{Log}(R_0^\top R_1)). \end{aligned} \quad (49)$$

Interpretation. Translations move linearly; orientations follow the geodesic in $\text{SO}(3)$. Controllers that require the whole interpolated path to be IK-feasible need these interior samples: endpoint reachability alone is insufficient.

8.3 Per-waypoint and path margins

Let A index angle candidates and S index Region A scenarios. After scoring,

$$\begin{aligned} g_{\ell,a} &= \mathcal{A}_S[F(T_{a,s}^{(\ell)})], \\ C_\ell &= \mathcal{A}_A[g_{\ell,a}], \end{aligned} \quad (50)$$

and for interior path samples C_m^{path} similarly. The scalar trajectory clearance is

$$C_{\text{traj}} = \text{softmin}_\tau \left[\{C_\ell\}_{\ell=1}^W \cup \{C_m^{\text{path}}\}_m \right] \quad (51)$$

(or CVaR / hard min). Optional SRS arm angle ψ enters as an outer softmax over ψ -conditioned fields when exposed.

8.4 Why the landscape updates with the trajectory

The network f_θ is trained once in the flange chart and then frozen. A trajectory is only a curve $x \mapsto (T_{\text{tcp}}(x), T_{\text{axis}}(x))$. Because:

1. c depends on the *relative* pose (7);
2. cone samples are expressed in the *current* TCP body frame (29);
3. aggregators (32)–(33) are memoryless functions of the current sample scores;

re-evaluating $C_{\text{task}}(s)$ or $C_{\text{traj}}(x)$ at a new path automatically yields a new IRD landscape and new gradients $\nabla_x C$ *without* modifying θ .

8.5 Typical optimisation objective

Ellipse-vessel demos minimise a hinge toward a conformal clearance target $m^\star$ plus regularisers:

$$\begin{aligned} \ell_{\text{IRD}} &= \frac{1}{W} \sum_\ell \text{softplus}\left(\frac{m^\star - C_\ell}{\sigma}\right) + \max_\ell \text{softplus}\left(\frac{m^\star - C_\ell}{\sigma}\right) - \kappa \bar{C}, \\ \ell &= w_{\text{IRD}} \ell_{\text{IRD}} + w_{\text{near}} \ell_{\text{nearest}} + w_{\text{cont}} \ell_{\text{continuity}} + \dots \end{aligned} \quad (52)$$

Interpretation. $\text{softplus}((m^\star - C)/\sigma)$ is a smooth one-sided penalty: inactive when $C \geq m^\star$, rising when clearance falls below target. The extra $-\kappa \bar{C}$ term continues to reward climbing toward IRD peaks even after the hinge is satisfied. Continuity penalties keep $\theta(s)$ and $y(s)$ smooth. Gradient descent on (52) is exactly “update the trajectory using conditional IRD gradients”.

9 End-to-end commutative diagram

$$\begin{aligned}
x &\xrightarrow{\text{kinematics / rail}} (T_{\text{tcp}}(x), T_{\text{axis}}(x)) \xrightarrow{\text{local samples}} \{T_k(x)\} \\
&\xrightarrow{(7)+(11)} \{c_k\} \xrightarrow{\tilde{f}} \{F_k\} \xrightarrow{A} C(x) \xrightarrow{\nabla} \nabla_x C
\end{aligned}$$

Proposition 1 (Single autograd tape). *If every arrow above is implemented with elementary differentiable maps, then $\nabla_x C$ exists almost everywhere and coincides with the analytic chain rule. Discrete IRD voxel lookups break the tape at the table index; the neural field does not.*

10 Scope, guarantees, and non-goals

- **Guidance, not certificate.** Held-out boundary errors are sub-millimetre when bracketed, but the field is an optimisation-direction operator. Hard feasibility still requires final multi-seed IK and collision checking.
- **Symmetry approximation.** Only J1 yaw is exact; residual probe-roll collision differences are rare and rejected downstream.
- **No branch / continuous ψ in the point field.** Offline labels are point reachability. Path residuals and continuous arm-angle selection live in the trajectory operator.
- **Not a full patient-trajectory planner by itself.** Region A / TaskCone / TrajectoryTask are query layers over the point field; application demos add geometric path parameterisations and loss terms.

11 Symbol table

Symbol	Meaning
$T_{\text{tcp}}, T_{\text{axis}}$	World TCP pose; world J1-axis pose
$c \in \mathbb{R}^9$	Yaw-invariant flange chart
$f_\theta, \tilde{f}$	Signed field; support-guarded field
F	World point score (20)
$C_A, C_{\text{task}}, C_{\text{set}}, C_{\text{traj}}$	Aggregated clearances
y	Rail coordinate
δ, η	Uncertainty sample; free (task) sample
τ	Softmin/softmax temperature
λ	Translation-rotation metric scale (21)
$\pi^{\min}, \pi^{\max}$	Softmin / softmax weights

12 References

1. N. Vahrenkamp, T. Asfour, and R. Dillmann, “Robot placement based on reachability inversion,” ICRA 2013.
2. M. Rudorfer, “RM4D: A Combined Reachability and Inverse Reachability Map for Common 6-/7-axis Robot Arms by Dimensionality Reduction to 4D,” ICRA 2025 / arXiv:2410.06968.
3. M. Murooka et al., “Learning Differentiable Reachability Maps for Optimization-based Humanoid Motion Generation,” Humanoids 2025 / arXiv:2508.11275.
4. S. L. Chiu, “Task Compatibility of Manipulator Postures,” International Journal of Robotics Research, 1988.

5. Internal design note: `ird_playground/docs/rm4d_signed_ird_design.md`.

A Code $\leftrightarrow$ equation map

Equation	Primary module
(8)	<code>region/operator.py::base_from_rail_torch</code>
(11)	<code>ird/canonical.py::canonical_flange_invariants_torch</code>
(20)	<code>neural/signed_field.py::score_world</code>
(32)–(33)	<code>region/operator.py</code> , <code>region/set_query.py</code>
(38)	<code>region/operator.py::RegionA</code>
(40)	<code>region/task_cone.py::TaskConeReachability</code>
(41)–(43)	<code>region/set_query.py::SetQueryOperator</code>
(49)–(51)	<code>region/trajectory_operator.py</code>
(24)–(28)	<code>neural/train_signed.py</code>
(52)	<code>experiments/ellipse_vessel_ird_demo.py</code>